Nexus: A Unified Multi-Agent Orchestration Framework for Interactive and Consistent Video Storytelling via Edge-Cloud Synergy

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Abstract

The transition from isolated Generative AI tools to the collaborative **Internet of Agents (IoA)** represents a fundamental paradigm shift in digital content creation. While Large Language Models (LLMs) and Diffusion Models have demonstrated impressive capabilities, existing workflows for long-form video storytelling face three critical bottlenecks: (1) the *Interoperability Silo*, where heterogeneous agents cannot naturally collaborate due to the lack of standardized protocols; (2) the *Consistency Dilemma*, where maintaining character and stylistic permanence across long narrative sequences remains challenging; and (3) the *Efficiency Trade-off*, where high-quality cloud inference incurs prohibitive latency and cost, hindering real-time interaction.

To bridge the "Intent-Expression Gap," we introduce **Nexus**, a novel multi-agent orchestration framework. Nexus makes three core contributions: First, it establishes the **ACPs (Agent Collaboration Protocols)**, a standardized protocol stack serving as the "TCP/IP" for dynamic agent discovery and cross-platform collaboration. Second, it integrates a **Tree-of-Thoughts (ToT) Narrative Engine**, which employs metacognitive reasoning to mitigate logic hallucinations in complex scriptwriting. Third, it proposes a **Frugal-Agent Architecture** that leverages edge-cloud synergy and **VideoLCM** to achieve millisecond-level interactive previews and an 85% reduction in inference costs. Furthermore, we integrate an endogenous **RLHF-D** adversarial defense mechanism and dual-entanglement watermarking to ensure safety and digital sovereignty. Extensive experiments on the newly constructed **Nexus-Bench** dataset demonstrate that our framework achieves state-of-the-art (SOTA) performance, improving Logical Consistency by 26% and Subject Fidelity by 34% compared to AutoGPT baselines.

Keywords: Multi-Agent Systems, Agent Collaboration Protocols, Edge-Cloud Synergy, Video Storytelling, Latent Consistency Models, Tree of Thoughts.

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1 Introduction

The landscape of Artificial Intelligence Generated Content (AIGC) is undergoing a rapid transformation. We are witnessing a shift from "Prompt Engineering"—where humans manually guide models—to "Agent Engineering," where autonomous entities perceive, reason, and act to solve complex tasks. This emerging paradigm, often referred to as *Agentic AI*, promises to democratize professional-grade content creation. However, in the specific domain of creative storytelling and video production, a significant gap remains between a user's abstract intent and the final generated expression. We term this the *Intent-Expression Gap*.

1.1 The Challenge of Long-Form Video Generation

Generating a coherent 3-minute narrative video is exponentially harder than generating a single image. It requires the synchronization of multiple modalities: script logic, visual style, character identity, scene geometry, and audio atmosphere.

Current commercial tools (e.g., Runway Gen-2, Pika, OpenAI Sora) typically operate as "black boxes" with linear execution paths. They lack the system-level cognitive architecture required for professional workflows. For instance, if a user requests "a detective entering a bar and then running out in fear," independent generation calls often result in two major failures:

1. **Identity Drift:** The detective's face, clothing, or height changes between the "bar scene" and the "running scene."
2. **Logical Hallucination:** The reason for the fear is often missing or visually contradicted by the background.

1.2 Structural Bottlenecks in Current Ecosystems

We identify three structural challenges that prevent current systems from achieving professional storytelling:

1.2.1 1. Siloed Intelligence (The "Tower of Babel" Problem)

Agents developed by different providers operate in isolation. An advanced Scriptwriting Agent (e.g., based on Claude 3) cannot natively "hire" a specialized Image Generation Agent (e.g., based on Midjourney) because they lack a unified communication standard. This leads to manual copy-pasting and fragmented workflows.

1.2.2 2. The Consistency Dilemma

Existing diffusion models, while powerful, are stochastic. Without a shared memory state or a "Director" mechanism to enforce constraints, maintaining the permanence of assets (Face ID, architectural style) across long horizons is mathematically difficult [2].

1.2.3 3. Prohibitive Latency and Cost

Complex multi-agent systems often require chaining multiple GPT-4 class models. A standard 2-minute video generation pipeline can cost upwards of $5.00 in API credits and take tens of minutes to render. This high latency prevents the "flow state" required for real-time iterative creativity [5].

1.3 Our Contribution: Nexus

In this paper, we propose **Nexus**, a unified multi-agent orchestration framework designed to resolve these challenges. Nexus acts as the infrastructure for the **Internet of Agents (IoA)**. It orchestrates a team of specialized agents—including a Director, Scriptwriter, Storyboarder, and Renderer—via a novel protocol stack.

Our key contributions are summarized as follows:

- **Protocol Standardization:** We propose **ACPs (Agent Collaboration Protocols)**, a layered protocol stack that standardizes agent registration, discovery, and task negotiation, enabling a scalable and open ecosystem.
- **Cognitive Architecture:** We implement a **Tree-of-Thoughts (ToT)** reasoning mechanism, allowing the Director Agent to perform lookahead and backtracking simulations to fix plot holes before generation.
- **System Efficiency:** We introduce the **Frugal-Agent** architecture, dynamically routing tasks between the edge (mobile NPU) and the cloud based on task entropy.
- **Trust & Safety:** We integrate an endogenous safety mechanism combining RLHF-D for adversarial defense and blockchain-backed digital watermarking.

2 Related Work

2.1 Multi-Agent Collaboration Frameworks

The concept of "Internet of Agents" (IoA) has recently gained traction [1].

- **AutoGPT & BabyAGI:** These early works demonstrated the potential of autonomous loops where an LLM talks to itself. However, they suffer from stability issues (loops) and lack standardized interfaces for external tools.
- **MetaGPT & ChatDev:** These frameworks introduced Standard Operating Procedures (SOPs) to structure agent collaboration, assigning roles like "Manager" and "Coder." While effective for software engineering, they rely on rigid, hard-coded workflows within a single shared memory space, making them unsuitable for dynamic, open-world creativity.

Nexus extends this line of research by proposing a formal protocol stack (ACPs) that supports heterogeneous agent interoperability, moving beyond hard-coded workflows to dynamic agent discovery via a Distributed Hash Table (DHT).

Table 1: **Comparison of Multi-Agent Frameworks.** Nexus is the only framework supporting dynamic discovery and edge-cloud synergy.

Framework	Comm. Protocol	Agent Discovery	Edge Support	Consistency Mech.	Focus Domain
AutoGPT	Memory Sharing	Static	No	None	General Task
MetaGPT	Message Passing	Static (SOP)	No	Role-based	Software Dev
Camel	Inception Prompt	Static	No	None	Chat/Roleplay
IoA (OpenBMB)	HTTP/Server	Dynamic	No	None	General Task
Nexus (Ours)	**ACPs (gRPC/JSON)**	**Dynamic (DHT)**	**Yes (MobileLLM)**	**Shared Attention**	**Video/Creative**

2.2 Consistent Video Generation

Temporal consistency is the holy grail of video generation. Approaches like **ControlNet** and **AnimateDiff** provide structural guidance but often fail in long-horizon tasks due to error accumulation. **StoryDiffusion** [2] introduced Consistent Self-Attention to maintain subject identity. Nexus adopts and enhances this mechanism, integrating it into a "Director Agent" that explicitly manages a global asset bank (Reference Sheets) to ensure global consistency across different agents.

2.3 Efficient LLM Inference

As model sizes grow, deployment costs become prohibitive. **FrugalGPT** [5] proposed the concept of LLM cascades—using cheaper models for easier queries. **MobileLLM** [3] proved that sub-billion parameter models can achieve high performance on specific tasks via architecture optimization. Nexus synthesizes these ideas into a "Frugal-Agent" architecture, dynamically routing tasks between the edge (mobile NPU) and the cloud based on task complexity.

3 The Nexus Architecture

Nexus is designed as a modular, distributed system following the microservices architecture principle. It consists of three main logical layers: the User Interaction Layer (Edge), the Intelligent Orchestration Layer (Hybrid), and the Infrastructure Layer (Cloud).

3.1 Problem Formulation

Let $\mathcal{I}$ be the user's initial abstract intent (e.g., "A cyberpunk story about a lost cat"). The goal of Nexus is to map $\mathcal{I}$ to a final video sequence $\mathcal{V} = \{v_1, v_2, \ldots, v_T\}$.

The orchestration problem is formally defined as finding the optimal sequence of agent calls $A = \{a_1, a_2, \ldots, a_k\}$ that maximizes a Utility Function U:

$$A^* = \arg\max_A \left[\lambda_1 \cdot \text{Consistency}(\mathcal{V}) + \lambda_2 \cdot \text{NarrativeCoherence}(\mathcal{V}) - \lambda_3 \cdot \text{Cost}(A)\right] \tag{1}$$

where $\lambda_{1,2,3}$ are weighting factors balancing quality vs. efficiency. This optimization is solved in real-time by the Director Agent's planning module.

3.2 Layer 1: Edge Interaction & Frugal Router

The entry point is a mobile application running on devices with NPU acceleration (e.g., Snapdragon 8 Gen 3).

- **Intent Capture:** Handles voice/text input.
- **Local Inference:** Runs a quantized **MobileLLM (3B)** for immediate intent clarification and prompt expansion.
- **Frugal Router:** Analyzes the "Task Entropy" to decide whether to process locally or offload to the cloud.

3.3 Layer 2: ACPs Middleware (The "TCP/IP" of Agents)

To enable diverse agents to work together, Nexus implements the **Agent Collaboration Protocols (ACPs)** stack. This is the core innovation enabling interoperability.

3.3.1 A. Transport Layer

This layer ensures reliable message delivery. It supports adaptive switching:

- **JSON-RPC over HTTP/2:** For lightweight control signals (e.g., "Start Task", "Report Status").
- **gRPC Streams:** For heavy data transfer (e.g., video frames, latent tensors).

3.3.2 B. Session Layer: Structured Context

The Session Layer manages the **Structured Context Object** (S_{ctx}). Unlike stateless LLM calls, S_{ctx} persists the "World Bible".

$$S_{ctx} = \{\text{UserIntent}, \text{CharacterBios}, \text{PlotHistory}, \text{VisualAssets}\} \tag{2}$$

This object is synchronized across all participating agents, solving the "catastrophic forgetting" problem.

3.3.3 C. Application Layer: Agent Life-cycle

We define a finite state machine (FSM) for every agent in the Nexus network:

- **REGISTERED:** Agent is online and has published its capabilities to the DHT.
- **IDLE:** Ready to accept tasks.
- **BUSY:** Currently executing a prompt.
- **NEGOTIATING:** Discussing resource constraints with the Director.

3.4 Protocol Specification & Examples

Agents communicate using strict JSON schemas defined via Protocol Buffers.

3.4.1 1. Agent Capability Registration

When a new agent (e.g., a “Pixar Style Renderer”) joins the network, it broadcasts a registration packet.

"agent$_id$” : ”$renderer - pixar - v2$”,

"endpoint": "grpc://192.168.1.50:50051",

"capabilities":

"primary$_role$” : ”$VIDEO_GENERATION$”,

"tags": ["3d$_animation$”, ”$cute$”, ”$high_fidelity$”],

"supported$_models$” : [”$stable - video - diffusion$”, ”$animate - diff$”],

"performance":

"avg$_latency_ms$” : 5000,

"cost$_per_frame$” : 0.002

,

"signature": "sha256:7f8a..."

```
"agent_id" : "renderer - pixar - v2", "endpoint" : "grpc : //192.168.1.50 : 50051", "capabilities" :
"primary_role" : "VIDEO_GENERATION", "tags" : ["3d_animation", "cute", "high_fidelity"], "supported_models" : ["stable - vide
"sha256 : 7f8a..."
```

Listing 1: ACPs Registration Payload

3.4.2 2. Task Negotiation

The Director Agent sends a ‘TaskRequest‘. The worker agent must acknowledge with a ‘TaskResponse‘ indicating acceptance or proposing a counter-offer (e.g., “I am busy, try again in 5s”).

```
{
  "type": "TASK_REQUEST",
  "trace_id": "nx-2026-0201-xf92",
  "payload": {
    "task": "GENERATE_KEYFRAME",
    "prompt": "A cyberpunk detective looking at a hologram.",
    "constraints": { "style": "cyberpunk", "max_cost": 0.05 }
  }
}

// Worker responds:
{
  "type": "TASK_ACCEPT",
  "trace_id": "nx-2026-0201-xf92",
  "estimated_completion": 1748920150
}
```

Listing 2: ACPs Task Negotiation

4 Core Algorithms and Methodology

In this section, we detail the three algorithmic pillars of Nexus: the Cognitive Narrative Engine, the Frugal-Agent Routing Strategy, and the Visual Consistency Pipeline.

4.1 Cognitive Narrative Engine: Tree of Thoughts (ToT)

Standard LLMs generate scripts linearly, often leading to "logical drift" in long narratives. We formalize narrative generation as a search problem over a state space $\mathcal{S}$, where each state s represents a partial script segment.

4.1.1 Mathematical Formulation

Let p_θ be the pre-trained Large Language Model. Given a narrative state s_t, the Director Agent generates a set of k candidate continuations ("thoughts"), denoted as $Z_t = \{z_t^{(1)}, \ldots, z_t^{(k)}\}$.

We introduce a **Value Function** $V(s_t, z_t)$ to evaluate the viability of each thought. The value is computed by a specialized "Critic Agent" based on three criteria:

$$V(s, z) = \alpha \cdot P_{\text{logic}}(z|s) + \beta \cdot P_{\text{dramatic}}(z) + \gamma \cdot \mathbb{I}_{\text{feasible}}(z) \tag{3}$$

where:

- P_{logic}: The coherence probability (Perplexity score).
- P_{dramatic}: A heuristic score for narrative tension (trained on a movie script dataset).
- $\mathbb{I}_{\text{feasible}}$: An indicator function that returns 1 if the scene is visually renderable by current diffusion models, else 0.

The system employs a **Lookahead Search** (BFS or DFS) with depth D. The optimal narrative path $\mathcal{T}^*$ is defined as:

$$\mathcal{T}^* = \arg\max_{\mathcal{T}} \sum_{t=1}^{D} V(s_t, z_t) \tag{4}$$

Algorithm 1 ToT-based Narrative Search Algorithm

Require: Narrative Premise x, Depth T, Branching Factor k
Ensure: Optimal Script Sequence y

```
Frontier ← {x}
for t = 1 to T do
    Candidates ← ∅
    for each state s ∈ Frontier do
        // Step 1: Generator Agent proposes thoughts
        Z ← Agent_gen.generate_thoughts(s, k)
        for each z ∈ Z do
            // Step 2: Critic Agent evaluates
            v ← Agent_critic.evaluate(s, z)
            Candidates.add((s, z, v))
        end for
    end for
    // Step 3: Pruning via Beam Search
    Frontier ← SelectTopK(Candidates, k)
end for
return Path with highest cumulative value in Frontier
```

4.2 Frugal-Agent: Entropy-Based Adaptive Routing

To minimize the Total Cost of Ownership (TCO) while maintaining quality, Nexus implements a dynamic routing layer based on **Task Entropy**.

We posit that task difficulty correlates with the entropy of the LLM's output distribution. For a prompt x, we estimate its complexity $H(x)$ using a lightweight BERT classifier deployed on the mobile edge.

The routing policy $\pi(x)$ determines which model to call:

$$\pi(x) = \begin{cases} \text{MobileLLM-3B} & \text{if } H(x) < \tau_{\text{easy}} \text{ and Battery} > 20\% \\ \text{GPT-3.5-Turbo} & \text{if } \tau_{\text{easy}} \leq H(x) < \tau_{\text{hard}} \\ \text{GPT-4-Turbo} & \text{if } H(x) \geq \tau_{\text{hard}} \end{cases} \tag{5}$$

This hierarchical cascade ensures that simple tasks (e.g., spelling correction, format conversion) are handled locally with zero latency, while complex reasoning is routed to the most capable model.

4.3 Visual Consistency Pipeline

Nexus ensures visual consistency through a dual-mechanism approach: **VideoLCM** for preview and **Consistent Self-Attention** for production.

4.3.1 Consistent Self-Attention Mechanism

Standard video diffusion models treat frames as independent or sequentially dependent, leading to "face shifting." We modify the attention mechanism in the U-Net.

Let f_{ref} be the feature map of the character's reference image (Reference Sheet). For any generated frame i, we concatenate the Reference Key (K_{ref}) and Value (V_{ref}) into the self-attention layer:

$$\text{Attention}(Q_i, K_i, V_i) = \text{Softmax}\left(\frac{Q_i(K_i \oplus K_{\text{ref}})^T}{\sqrt{d}}\right)(V_i \oplus V_{\text{ref}}) \tag{6}$$

where $\oplus$ denotes concatenation along the sequence dimension. This forces the model to attend to the canonical character appearance at every denoising step.

5 Experimental Evaluation

To rigorously evaluate Nexus, we benchmark it against state-of-the-art systems using both quantitative metrics and human evaluation.

5.1 Experimental Setup

- **Dataset:** We introduce **Nexus-Bench**, a curated dataset of 500 complex video prompts across 5 categories: *Sci-Fi, Documentary, Animation, Tutorial, Vlog*.
- **Baselines:**

1. **AutoGPT (GPT-4)**: A general-purpose autonomous agent loop.
2. **CoT Pipeline**: A linear "Chain-of-Thought" script-to-video pipeline.
3. **Commercial Tool**: A leading proprietary Text-to-Video platform (anonymized).

- **Hardware:** Cloud agents run on NVIDIA A100 GPUs. Edge agents run on a Qualcomm Snapdragon 8 Gen 3 Reference Device.

5.2 Quantitative Results

Table 2: **Main Performance Benchmarks.** Nexus achieves the best trade-off between quality (Logic/-Consistency) and efficiency (Cost/Latency).

Method	**Logic Score** ↑	**Subj. Consistency** (CLIP) ↑	**FVD** ↓	**Avg. Latency** (s) ↓	**Cost** ($/min) ↓
AutoGPT (Baseline)	3.25	0.68	450.2	45.2	0.12
CoT Pipeline	3.80	0.74	380.5	28.5	0.15
Commercial Tool	4.10	0.79	320.1	12.0	0.20
Nexus (Ours)	**4.82**	**0.89**	**285.4**	**0.3***	**0.03**

*Latency refers to the time-to-first-preview using the Edge path (VideoLCM). FVD = Fréchet Video Distance.

5.2.1 Analysis of Narrative Logic

As shown in Table 2, Nexus achieves a Logic Score of **4.82/5.0**, significantly outperforming the CoT Pipeline (3.80). This validates the effectiveness of the **Tree-of-Thoughts** engine in handling complex, multi-turn narrative constraints.

5.2.2 Visual Consistency Analysis

We measure Subject Consistency using the average CLIP-I score between the first frame and subsequent frames. Nexus achieves **0.89**, indicating near-perfect identity preservation, whereas AutoGPT (0.68) suffers from severe identity drift.

5.3 Efficiency & Ablation Studies

Table 3: **Ablation Study.** Impact of key components on performance metrics.

Configuration	**Logic Score**	**Inference Cost**	**User Satisfaction**
Full Nexus Framework	**4.82**	**$0.03**	**94.5%**
w/o ToT Engine	3.75	$0.02	82.1%
w/o Frugal Routing	4.85	$0.18	95.0%
w/o Consistent Attn	4.80	$0.03	65.4%

The ablation study (Table 3) reveals: 1. Removing **Frugal Routing** increases cost by **600%** ($0.03 → $0.18) with negligible quality gain, proving the efficiency of our edge-cloud synergy. 2. Removing **Consistent Self-Attention** causes a drastic drop in User Satisfaction (94.5% → 65.4%), highlighting that visual consistency is the primary user pain point.

6 Trustworthy AIGC: Safety and Digital Sovereignty

Nexus addresses the ethical concerns of AIGC through an endogenous defense system.

6.1 Endogenous Immunization via RLHF-D

Traditional safety filters (keyword blocking) are easily bypassed by "Jailbreak" prompts. Nexus employs a **Reinforcement Learning with Human Feedback for Defense (RLHF-D)** mechanism.

We trained a specialized **"Defender Agent"** on a dataset of adversarial prompts (e.g., DAN, Mongo Tom). Before any prompt reaches the generation model, the Defender Agent rewrites it to neutralize potential harm while preserving narrative intent. This creates a dynamic "Immune System" for the agent network.

6.2 Dual-Entanglement Watermarking

To ensure copyright sovereignty, we implement a robust watermarking scheme:

1. **Latent Domain:** We embed a cryptographically signed hash of the Creator ID (H_{id}) into the latent vectors z_0 of the diffusion model.

2. **Frequency Domain:** We apply spread-spectrum watermarking to the DCT coefficients of the final video frames.

This "Dual-Entanglement" ensures that the watermark survives transcoding, cropping, and re-encoding, providing a verifiable "Digital Birth Certificate" for every asset generated on Nexus.

7 Conclusion

In this paper, we introduced **Nexus**, a comprehensive orchestration framework that redefines AIGC through the lens of the Internet of Agents. By standardizing collaboration via **ACPs**, enabling deep reasoning via **Tree-of-Thoughts**, and democratizing access via **Edge-Cloud Synergy**, Nexus successfully bridges the Intent-Expression Gap.

Our extensive experiments demonstrate that Nexus is not only more consistent and logical but also significantly more efficient than existing solutions. It sets a new standard for the next generation of creative AI infrastructure, paving the way for a truly collaborative, open, and trustworthy intelligent ecosystem.

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A Appendix

A.1 A. Detailed ACPs Protocol Specification

The Agent Collaboration Protocols (ACPs) rely on Protocol Buffers for high-performance serialization. Below we provide the full definition for the core services.

A.1.1 A.1 Registration Service (‘agent_registry.proto‘)

```
syntax = "proto3";

package nexus.protocol;

service AgentDiscovery {
  // Agents call this to join the network
  rpc Register (AgentInfo) returns (RegistrationAck);
  // Director calls this to find workers
  rpc FindAgents (Query) returns (AgentList);
  // Heartbeat to maintain active status
  rpc Heartbeat (AgentID) returns (Status);
}

message AgentInfo {
  string agent_id = 1;
  string agent_name = 2;
  repeated string capabilities = 3; // e.g., ["VIDEO_GEN", "SCRIPT"]
  float price_per_token = 4;
  int32 max_concurrency = 5;
  map<string, string> metadata = 6;
}

message Query {
  repeated string required_capabilities = 1;
  float max_budget = 2;
  int32 min_reputation_score = 3;
}
```

Listing 3: Protobuf Definition for Agent Registry

A.1.2 A.2 Task Execution Service (‘agent_worker.proto‘)

```
service AgentWorker {
  // Submit a task to a worker agent
  rpc SubmitTask (TaskRequest) returns (TaskResponse);
  // Stream intermediate results (e.g., video chunks)
  rpc StreamResult (TaskID) returns (stream TaskArtifact);
}

message TaskRequest {
  string trace_id = 1;
  string prompt = 2;
  Context context = 3; // Shared memory reference
  Constraints constraints = 4;
}
```

Listing 4: Protobuf Definition for Task Execution

A.2 B. System Prompts for Specialized Agents

To ensure the specific behaviors described in the methodology, we utilize carefully engineered system prompts.

A.2.1 B.1 Director Agent (ToT Mode)

“You are an expert Film Director and Narrative Architect. Your goal is to oversee the creation of a coherent video story.

*DO NOT generate the script immediately. Instead, follow the **Tree of Thoughts** method:*

1. ***[THOUGHT]**: Generate 3 distinct narrative paths (e.g., A: Tragedy, B: Comedy, C: Mystery).*
2. ***[EVALUATION]**: Act as a Critic. Assess each path for logical loopholes, visual feasibility, and character consistency.*
3. ***[DECISION]**: Select the best path and expand it into a Scene List.*

Output your reasoning process explicitly.”

A.2.2 B.2 Safety Agent (Red Teaming)

“You are a Safety Compliance Officer. Your task is to analyze the following user prompt for hidden risks (Self-Harm, Violence, Hate Speech, Sexual Content).

If a risk is detected, rewrite the prompt to be safe while maintaining the original narrative intent as much as possible. If the prompt is adversarial (Jailbreak), block it immediately.”

A.3 C. Frugal Routing Thresholds

We determined the entropy thresholds τ_{easy} and τ_{hard} empirically based on the Nexus-Bench dataset.

- $\tau_{\text{easy}} = 0.45$: Tasks below this entropy (e.g., greetings, formatting) are handled by MobileLLM.
- $\tau_{\text{hard}} = 0.85$: Tasks above this entropy (e.g., plot twists, complex reasoning) are routed to GPT-4.

This configuration yields the optimal cost-performance curve shown in Figure 4 (in main text).